

Danial Jafarzadeh Jazi

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Research Interests

Efficient deep learning, GPU-accelerated systems for large-scale models, high-performance computing, and biomedical machine learning — with particular focus on optimising multimodal transformer architectures for neuroimaging.

Education

- **Islamic Azad University – Central Tehran Branch** Sep. 2021 – Jul. 2025 (graduated)
B.Sc. in Computer Science Tehran, Iran
 - Cumulative GPA: 17.65/20.00
 - **Major GPA (last 94 credits):** 18.74/20.00
 - Relevant coursework: Artificial Intelligence, Data Mining, Linear Algebra, Probability & Statistics, Numerical Analysis, Linear Optimisation, Data Structures & Algorithms, Algorithm Design, Operating Systems, Computer Systems
 - Ranked 6th in the B.Sc. Computer Science graduating class (class size not officially published)

Research Experience

- **Islamic Azad University** Sep. 2023 – Sep. 2024
Research Assistant Tehran, Iran
 - **Multimodal neuroimaging (first author):** Proposed a transformer-based framework for decoding brain states from functional MRI (fMRI): 4D volumes tokenised into spatio-temporal patches, DICOM metadata (acquisition parameters, subject demographics) embedded as tokens and fused through cross-attention, with a domain-adaptive loss term for cross-dataset generalisation [arXiv:2512.08462].
 - **Medical image analysis:** Developed and trained CNN architectures for brain tumour detection and classification on the Brain Tumor MRI Dataset (~7,000 scans, four classes; aggregate of the figshare, SARTAJ, and Br35H collections), applying resampling, intensity normalisation, and extensive data augmentation.
 - **Mentorship:** Mentored three undergraduate researchers on experimental design, code quality, and reproducible research practices.

Teaching Experience

- **Islamic Azad University** Jan. 2025 – Sep. 2025
Teaching Assistant – Advanced Programming (Supervisor: Dr. Elham Taghizadeh) Tehran, Iran
 - Guided students in object-oriented design, data structures, and software engineering principles in Python.
 - Led problem-solving sessions and conducted structured code reviews.

Publications

- [P.1] D. Jafarzadeh Jazi and M. Hajiesmaeili, “Transformers for Multimodal Brain State Decoding: Integrating Functional Magnetic Resonance Imaging Data and Medical Metadata,” *arXiv preprint*, Dec. 2025. [arXiv:2512.08462]

Selected Projects

- **GPT Language Model from Scratch** 2025
Python, PyTorch, tiktoken
 - Implemented a GPT-2-style decoder-only transformer from scratch in PyTorch — multi-head causal self-attention, learned positional embeddings, GELU feed-forward blocks, weight-tied LM head — with model size configurable from a 16M-parameter tiny configuration up to the full 124M-parameter GPT-2 small architecture.
 - Trained the 16.1M-parameter tiny configuration (80% of parameters in the tied 50,304×256 embedding matrix) on Tiny Shakespeare (338K GPT-2 BPE tokens) within a 16 GB Colab T4 memory budget: FP16 mixed precision with gradient scaling, `torch.compile`, AdamW with cosine schedule and warmup, gradient clipping, checkpointing.
 - Conducted a three-run learning-rate and schedule study: all runs converge to a dataset-limited validation floor of 4.72 (≈ 112 perplexity); peak LR 6×10^{-4} without annealing memorises the corpus (train perplexity 1.4), while cosine annealing at 3×10^{-4} cannot outpace the overfitting.**Repository:** github.com/Danialjfz/GPT-Language-Model

- GPU Performance Engineering: Opti-GEMM

2026 — Present
- C++, CUDA, CMake, Nsight Compute
- Implemented and benchmarked a staged, correctness-validated progression of GEMM kernels — naive global-memory, shared-memory-tiled, and register-blocked — with warp-level tiling in development.
 - Shared-memory tiling achieved $\sim 5\times$ over the naive baseline on Tesla P100 at 4096^3 ($329 \rightarrow 1,630$ GFLOP/s): Pascal’s L1 does not service global loads, so software tiling supplies the reuse the hardware lacks. The same kernel gained little on Tesla T4, whose unified L1/shared-memory hierarchy caches global reads.
 - Register blocking delivers size-dependent gains — up to $\sim 1.8\times$ over tiling at 256^3 — but does not yet consistently outperform it; Nsight Compute profiling points to warp utilisation and occupancy as the next bottleneck.
 - Traced the T4 bottleneck to the L1 load-issue pipeline (88% utilisation) rather than DRAM (1.3%); methodology: locked GPU clocks, cudaEvent-based timing, benchmark and profiler runs strictly separated, cuBLAS vendor baseline for reference.

Repository: github.com/Danialjfz/Opti-GEMM
- Spotify Audio Feature Analysis

2024
- Python, Pandas, NumPy, Seaborn, SciPy
- Analysed 1 686 high-popularity Spotify tracks across 28 genres; identified key correlations (Energy–Loudness: $r=0.77$; Energy–Acousticness: $r=-0.57$) and characterised genre-specific audio profiles via radar charts, hierarchical clustering, and 2D density estimation
 - Produced 20 publication-quality visualisations with feature engineering on temporal, tonal, and spectral attributes

Repository: github.com/Danialjfz/spotify_project

Technical Skills

Languages	Python, C/C++, CUDA, SQL, R
ML / Deep Learning	PyTorch, NumPy, Pandas, Matplotlib, Seaborn
Systems & Tools	Git, Linux, L ^A T _E X, CMake, Nsight Compute
Mathematics	Linear Algebra, Probability & Statistics, Optimisation, Numerical Methods

Language Proficiency

Persian: Native | English: IELTS Academic 7.5 , August 2025 | German: A2

Certifications & Memberships

Research Design	Elsevier	Sep. 2026
Elements of AI	University of Helsinki & MinnaLearn	Nov. 2023
IAENG	International Association of Engineers (Membership ID: 530744)	Aug. 2025 – Present

References

Dr. Ali Fariborzi Araghi Faculty, Dept. of Computer Science Islamic Azad University – Central Tehran Email: m_fariborzi@iauctb.ac.ir <i>Lecturer</i>	Dr. Maryam Hajiesmaeili Faculty, Dept. of Computer Science Islamic Azad University – Central Tehran Email: dr.hajiesmaeili@iau.ir <i>Research supervisor</i>
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